

Discrete Drainage Dynamics: A Bounded, Conservative Local Rule for Newtonian-Like Potentials

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Abstract

We define a deterministic, parallel, local update rule on a discrete graph. Each node carries a non-negative reserve $R_i \geq 0$, with rest value R_0 at equilibrium. Each oriented link carries a non-negative directional flux. The rule alternates a node update and a link update, with the outgoing fluxes from a node rate-limited so that the reserve cannot become negative. The flux on a link is written in a general form involving a *mobility function* $\mu(R)$ which encodes how the source node's reserve modulates its emission rate. The present paper adopts the constant-mobility baseline $\mu \equiv 1$, the simplest choice consistent with the substrate's locality and conservation properties. In this baseline, far from the non-negativity floor, the rule's steady state coincides with the discrete Poisson equation, and we measure on the lattice that the static profile around a point source follows a $1/r$ law with the continuum coefficient at the percent level. Companion papers study other mobility prescriptions and their non-linear consequences.

1 What this paper claims

We make three claims.

1. We define a deterministic, parallel, two-step local rule on a discrete graph. The reserve is bounded below by zero, R_0 is the rest value, and the flux on a link involves a mobility function $\mu(R)$ that we leave as a structural choice of the framework. The present paper studies the constant-mobility baseline $\mu \equiv 1$.
2. Run as a stationary linearised simulation with a point source on a cubic lattice, the rule (with $\mu \equiv 1$) produces a radial profile compatible with $1/r$ in the bulk, matching the discrete continuum prediction at the percent level across $L \in \{64, 128, 160\}$.
3. The internal flux-exchange rule preserves total reserve exactly. Non-negativity of the reserve is enforced by rate-limiting the outgoing fluxes and external sinks of a node in proportion. This is the structural feature that distinguishes the rule from a plain discrete heat equation.

We do not claim the following. We do not derive Newton's gravitational constant, general relativity, or any property of the Standard Model. We do not derive the mobility function $\mu(R)$; the constant-mobility baseline of this paper is a choice within a family of possibilities. We do not assert that the $1/r$ result is itself novel: it is not. What is novel is the rule, the mobility framing, and the structural features that make the rule non-trivial in the regimes explored in the companion papers.

2 The picture, in plain words

Imagine a fine three-dimensional grid. At each grid point sits a reservoir holding a non-negative amount of stuff, with rest level R_0 . Reservoirs can be excited above R_0 or drained below it, but they cannot go below zero. Between neighbours run pipes that carry flow from the fuller reservoir to the emptier one, at a rate proportional to the level difference and optionally modulated by the source level. A reservoir cannot deliver more than it holds.

If we start at rest level everywhere, drain one point, and let it settle, with $\mu \equiv 1$ the bulk pattern is the textbook discrete Poisson solution, $\delta(r) = R_0 - R(r)$ falling as $1/r$. Other mobility choices give different non-linear regimes (companion papers).

3 The substrate

The substrate is a graph with at least six neighbours per node and statistical isotropy. We use cubic lattice in simulations. Each node i carries $R_i \geq 0$. Each oriented link carries non-negative directional fluxes $F_{i \rightarrow j}$.

The full rule of the framework also assigns angular quadratures (T_i, I_i) with $T_i^2 + I_i^2 \leq R_i$ and link phases ψ_{ij} , used in the gauge sector (Papers V–VI). They do not enter the present paper.

Parameter	Role	Used in this paper
κ	drainage coefficient	yes
α	flux propagation coefficient	yes
R_0	reserve rest value	yes
$\mu(R)$	mobility function	yes ($\mu \equiv 1$)
J	link relaxation coupling	no (deferred)

Table 1: Parameters of the rule. The mobility $\mu(R)$ is a structural choice: this paper uses $\mu \equiv 1$, Paper II uses $\mu(R) = R/R_0$.

4 The two-step tick

The rule decomposes into a Link Update followed by a Node Update, alternating each tick. Locality is preserved (each update reads only nearest neighbours). Two-step alternation is a standard local lattice construction.

4.1 Link Update

For each oriented link (i, j) in parallel, compute a desired flux,

$$F_{i \rightarrow j}^{\text{des}} = \alpha \mu(R_i) (R_i - R_j)_+, \quad (1)$$

where $\mu(R)$ is a non-negative mobility function with $\mu(R_0) = 1$. We do not derive $\mu(R)$ from a deeper principle; it is a structural choice of the rule.

Constant-mobility baseline of this paper. We adopt

$$\mu(R) \equiv 1, \quad (2)$$

for which Eq. (1) reduces to $F_{i \rightarrow j}^{\text{des}} = \alpha(R_i - R_j)_+$, and the linearised steady state is the discrete Poisson equation.

Other mobility choices in companion papers. The reserve-proportional mobility $\mu(R) = R/R_0$, motivated by local bandwidth limitation at the source, is studied in Paper II and produces $\nabla^2(R^2) = 0$ in the continuum limit, yielding a Schwarzschild-form profile. Other choices may be explored elsewhere.

Each node computes a rate-limiter

$$A_i = R_i + \tau S_i^{\text{in}}, \quad S_i^{\text{in}} = \sum_j F_{j \rightarrow i}^{\text{des}}, \quad D_i = \tau O_i^{\text{des}} + \tau \kappa E_i, \quad O_i^{\text{des}} = \sum_j F_{i \rightarrow j}^{\text{des}}, \quad (3)$$

$$\beta_i = \begin{cases} 1, & D_i = 0, \\ \min(1, A_i/D_i), & D_i > 0, \end{cases} \quad (4)$$

applied identically to outflows and sink: $F_{i \rightarrow j} = \beta_i F_{i \rightarrow j}^{\text{des}}$, $E_i^{\text{real}} = \beta_i E_i$.

4.2 Node Update

$$R_i \leftarrow R_i + \tau S_i^{\text{in}} - \beta_i \tau O_i^{\text{des}} - \beta_i \tau \kappa E_i. \quad (5)$$

By construction, $R_i \geq 0$ at every tick.

5 Conservation

Internal flux exchange conserves total reserve exactly: realised fluxes enter Node Update of j as $+\beta_i F_{i \rightarrow j}^{\text{des}}$ and Node Update of i as $-\beta_i F_{i \rightarrow j}^{\text{des}}$. External sinks $E_i^{\text{real}} = \beta_i E_i$ are explicit bookkeeping. The closed system network-plus-sink-reservoir is conservative.

6 Simulation: from rule to $1/r$

6.1 Derivation of the Poisson limit (constant mobility)

With $\mu \equiv 1$ and $\beta_i = 1$, the two directional fluxes satisfy $F_{i \rightarrow j} - F_{j \rightarrow i} = \alpha(R_i - R_j)$, so $S_i = \alpha \Delta R_i$. Stationary regime gives $\alpha \Delta R_i = \kappa E_i$, and with $\delta_i = R_0 - R_i$:

$$-\Delta \delta_i = M \delta_{\text{Kr}}(i, i_0), \quad M = \kappa E_0 / \alpha. \quad (6)$$

Continuum solution $\delta(r) = M/(4\pi r)$.

6.2 Numerical check

L	A_{fit}	rel. bias	precision	N_{iter}
64	0.07878	-1.01%	7.0×10^{-5}	5 000
128	0.07866	-1.16%	5.7×10^{-6}	10 000
160	0.07886	-0.90%	3.7×10^{-6}	30 000
$A_{\text{continuum}} = M/(4\pi)$		0.07958		

Table 2: Newton-fit on cubic lattices, $M = 1$, Dirichlet boundary, $\mu \equiv 1$.

6.3 Robustness: spectral dimension

Spectral dimension converges to 3 as L grows (cubic $d_s - 3$ minima: 0.129, 0.054, 0.023 at $L = 32, 64, 128$). The gravitational sector is Newtonian in the IR limit.

6.4 Code and data

<https://github.com/stanislasdewavrin/DDD-Paper-1-Foundations>. Pure-numpy.

7 Calibration vs. prediction

Calibration. The mappings $\delta \leftrightarrow \Phi$ and $E \leftrightarrow m$ are external. The lattice unit and κ are fixed by demanding A_{fit} matches Newton’s G .

Prediction. Within that family, the rule with $\mu \equiv 1$ predicts a static $1/r$ potential.

8 What the model is not, and what it does not do

We do not derive general relativity, Newton’s G , the speed of light c , or any property of the Standard Model.

We do not derive the mobility function $\mu(R)$. The present paper studies the constant-mobility baseline $\mu \equiv 1$, for which the linearised steady state is the discrete Poisson equation. Different mobility choices lead to different non-linear steady-state equations; the constant-mobility choice is the minimal Newtonian baseline. Companion papers study specific reserve-dependent mobility prescriptions, in particular the source-side bandwidth-throttled choice $\mu(R) = R/R_0$ in Paper II.

We have not shown the rule is unique.

9 Open questions

Why six neighbours? Minimal cubic coordination, not a fundamental requirement.

Why this order, Node then Link? Microscopic breaking of time-reversal; not derived.

What sets R_0 ? Only structural scale of the rule; not derived.

What sets the mobility $\mu(R)$? The constant-mobility baseline of this paper is the simplest choice. Other choices (notably the bandwidth-throttled $\mu(R) = R/R_0$ of Paper II) yield qualitatively different non-linear regimes. A microscopic argument that selects a specific mobility from the rule would close an important gap.

Is the rule unique? See Sec. 8.

What sits below the substrate? Beyond the present paper.

10 Outlook

1. Paper II — weak-field gravitational behaviour with the source-side bandwidth-throttled mobility $\mu(R) = R/R_0$, yielding the Schwarzschild time-dilation factor. *Linearised plus saturation; partial.*
2. Paper III — discrete-time response and inertial effects. *Speculative.*
3. Paper IV — predictions and tests of the gravitational sector. *Mixed.*
4. Paper V — gauge sector and emergent electromagnetism. *Speculative.*
5. Further notes (Papers VI and onward) address coupling, cosmology, quantum phenomenology, particle structure. *Increasingly speculative.*

The present paper, Paper I, claims only the rule itself (with constant-mobility baseline) and the linearised $1/r$ verification. The rest is what one might do with such a rule.

References